

9. Iron is an example of a transition element. It has two main oxidation states in its compounds.

- (a) Write the electronic structure of an Fe^{2+} ion and use it to show why iron is classed as a transition element. [2]

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- (b) Explain why iron has more than one common oxidation state in its compounds. [1]

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- (c) In very acidic solutions, Fe^{3+} forms $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$.

Draw the structure of the $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ ion. [1]



- (d) In solutions with pH values between 1 and 3, the following equilibrium occurs.



The concentrations of $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ and $[\text{Fe}(\text{H}_2\text{O})_5(\text{OH})]^{2+}$ present can be studied using colorimetry.

- (i) Explain why the complex ions $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ and $[\text{Fe}(\text{H}_2\text{O})_5(\text{OH})]^{2+}$ are not the same colour. [2]

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- (ii) Suggest how you would select an appropriate wavelength to find the concentration of $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ in the equilibrium mixture. [1]

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- (iii) Write an expression for the equilibrium constant K_c for this equilibrium, giving its unit. [2]

Unit



- (iv) A known mass of iron(III) chloride, $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ (M_r 270.4) is dissolved in 1.00 dm^3 of dilute hydrochloric acid. At equilibrium the pH of the solution is 1.55 and the concentration of $[\text{Fe}(\text{H}_2\text{O})_5(\text{OH})]^{2+}$ is $0.103 \text{ mol dm}^{-3}$. The numerical value of K_c under these conditions is 4.03×10^{-3} .

Find the mass of $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ used to make the solution.

[4]

Mass = g



(e) At higher pH values, solutions containing hydrated Fe^{3+} ions form a precipitate.

(i) Give the formula of the precipitate and its colour. [1]

Formula

Colour

(ii) Aqueous iron(II) compounds form a precipitate of a different colour with aqueous sodium hydroxide. In a test tube, the precipitate can change colour on standing. Use the standard electrode potential values below to explain this change. [2]

	Standard electrode potential, E^θ / V
$\text{Fe}^{3+} + \text{e}^- \rightleftharpoons \text{Fe}^{2+}$	+0.77
$\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightleftharpoons 2\text{H}_2\text{O}$	+1.23

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(iii) A student states that it would be incorrect to use the standard electrode potential for the oxygen half-cell as the solution will be alkaline. State, giving a reason, the effect an alkaline solution will have on the value of the electrode potential of the oxygen half-cell. [2]

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(f) In a blast furnace Fe_2O_3 is reduced by CO to form iron metal.

(i) Write an equation for this process.

[1]

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(ii) Explain why CO is a good reducing agent.

[1]

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